The boundary diagonal in type B:
$\mathrm{ord}_S \widetilde{A}^B_{\mathbf{I},\mathbf{I}} = [\, m \in \mathbf{I}\,]$, and why the order is one

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Abstract

I close G3 of `2026-10-19-c2-sa-divisors-full-q.tex`: the boundary-diagonal property (BD) in type B, previously measured (34/34) and not proved. The proof is three lemmas and is uniform in type; the statement it proves is strictly more general than the one asked for:

$$\mathrm{ord}_{S^\varepsilon_m} \widetilde{A}^X_{\mathbf{I},\mathbf{I}} \;=\; [\, m \in \mathbf{I}\,] \cdot \#\{\, a > 0 \;:\; a\, t_m \in \Delta^+(X_N) \text{ and } \varepsilon^a = 1 \,\}.$$

Three consequences. (1) **G3 is closed**, so everything in §§5–7 of the 10-19 paper is now proved in *both* types. (2) **G4 is derived rather than measured**: type B has no $\varepsilon = -1$ component because its boundary root is t_m and not $2t_m$, and I give the exact failure pattern. (3) "Order exactly one" *is* reducedness of the ambient root system — in the non-reduced BC_N, where t_m and $2t_m$ are both roots, the same computation returns 2. I also **retract a hypothesis of my own statement**: (BD) needs no genericity condition. Pre-registration **P1 came back SPLIT**, and the split is sharp: the two halves of (BD) carry *complementary* hypotheses, each inert in the other half.

1 What was open, and what the gap actually was

The 10-19 paper fixes $m \in [N]$ and ε with $\varepsilon^h = 1$, sets $S = S^\varepsilon_m := \{w_m = \varepsilon\}$, and uses two properties of Motegi's normalised weight functions $\widetilde{A}^X_{\mathbf{M},\mathbf{I}} := \widetilde{WX}_{N,n,r}(\mathbf{w_M}|\mathbf{w}|\mathbf{I})$ of `arXiv:2610.15344`: (BC), the boundary collapse, and

$$\textbf{(BD)} \qquad \mathrm{ord}_S \widetilde{A}^X_{\mathbf{I},\mathbf{I}} = [\, m \in \mathbf{I}\,] \quad \text{for every } \mathbf{I}.$$

(BC) is proved in both types. (BD) is proved in type C (inside `prop:short` of `2026-10-17-c2`) and was recorded as **G3**, "measured, not proved", in type B — 34/34 at $N \le 3$. Everything in §§5–7 was accordingly graded *proved in type C, conditional on G3 in type B*.

Remark 1 (the gap was a reading gap, and the locator says so). The G3 note points at "the type-B diagonal is `normalizationWB` (L1386)". L1386 is the *unexpanded* statement $\prod_{\alpha\in\mathrm{Inv}(\lambda)}(1-e^{-\alpha})\prod_{\alpha\in T(\lambda)}(1-qe^{+\alpha})$ — which is *verbatim the same formula* as the type-C one at L1078. The explicit type-B expansion, with the root system written out, is at **L1305–1335**, some eighty lines *earlier*, in a subsection whose opening sentence is "The proofs of the type B statements are essentially the same as the corresponding type C arguments… For brevity, we omit these proofs" (L1281). I had never opened L1305–1335. My (M4) instead carried a *parameterised guess*, "the boundary term $(1-w_i^{-h})$ occurs"; the guess was right, which is exactly why it survived. Once the region is read, the proof below is three lemmas. This is the same shape as the broken-locator finding of 2026-10-18: a citation that lands next to the content rather than on it is evidence that the neighbourhood was never read, and the repair is to read around it, not to patch the reference.

2 Setup, read first-hand from the source

Write $h = h_X$ for the *boundary coefficient*: $h_B = 1$, $h_C = 2$. From L1288–1292 (type B) and L960–964 (type C),

$$\Delta^+(X_N) = \{\, t_k \pm t_j \;:\; 1 \le j < k \le N \,\} \;\cup\; \{\, h\, t_j \;:\; 1 \le j \le N \,\},$$

so the two systems differ *only* in the boundary root, t_j versus $2t_j$. Set $e^{t_j} := w_j$, extended multiplicatively, so $e^{\sum_j a_j t_j} = \prod_j w_j^{a_j}$.

For $\mathbf{I} \subseteq [N]$ with associated strict partition $\lambda = \lambda(\mathbf{I})$, Motegi computes the inversion and ascent sets explicitly (L998–1003, L1013–1019 type C; L1305–1312, L1322–1328 type B). In both types, *with the same combinatorics and only h changing*:

$$\mathrm{Inv}(\lambda) = \{ht_k : k \in \mathbf{I}\} \cup \{t_j + t_k : j < k,\ j,k \in \mathbf{I}\} \cup \{t_k - t_j : j < k,\ j \notin \mathbf{I},\ k \in \mathbf{I}\}, \tag{1}$$

$$T(\lambda) = \{ht_k : k \notin \mathbf{I}\} \cup \{t_j + t_k : j < k,\ j,k \notin \mathbf{I}\} \cup \{t_k - t_j : j < k,\ j \in \mathbf{I},\ k \notin \mathbf{I}\}, \tag{2}$$

and the diagonal value is (`normalizationWC` L1078, `normalizationWB` L1386)

$$\widetilde{A}^X_{\mathbf{I},\mathbf{I}} = \prod_{\alpha\in\mathrm{Inv}(\lambda)} \left(1 - e^{-\alpha}\right) \prod_{\alpha\in T(\lambda)} \left(1 - q\, e^{+\alpha}\right). \tag{3}$$

Definition 2 (the valuation)**.** Let $F = \mathbb{C}(q)(w_1, \ldots, w_N)$ and $R = \mathbb{C}(q)[w^{\pm}]$. Fix $m \in [N]$ and $\varepsilon \in \mathbb{C}^\times$. Since $R/(w_m - \varepsilon) \cong \mathbb{C}(q)[w_j^{\pm} : j \neq m]$ is a domain (here $\varepsilon \neq 0$ is used), $(w_m - \varepsilon)$ is a prime of R; let $\mathcal{O} \subset F$ be the local ring, $t := w_m - \varepsilon$ the uniformiser, and $k = \mathcal{O}/(t) \cong \mathbb{C}(q)(w_j : j \neq m)$ the residue field. Write ord_S for the associated divisorial valuation on $F^\times$ and $\bar{x} \in k$ for the residue of $x \in \mathcal{O}$.

The one structural fact used repeatedly: ord_S is a valuation, so it is *additive* on products, and $\mathrm{ord}_S\, x = 0$ exactly when $x \in \mathcal{O}^\times$, i.e. when $\bar{x} \neq 0$ in k.

3 The unit criterion

Everything in (3) is of the form $1 - c\,M$ with M a Laurent monomial. One lemma handles all of them.

Lemma 3 (unit criterion)**.** *Let $M = \prod_j w_j^{a_j}$ be a Laurent monomial and $c \in \mathbb{C}(q)^\times$. Then $1 - cM \in \mathcal{O}$, and*

(i) $\operatorname{ord}_S(1-cM) \geq 1 \iff a_j = 0$ *for all* $j \neq m$ *and* $c\,\varepsilon^{a_m} = 1$;

(ii) *if moreover* $a := a_m \neq 0$, *then* $\operatorname{ord}_S(1-cM) = 1$ exactly, *and the cofactor of* t *at the generic point of* S *is* $a\,\varepsilon^{a-1}$.

Proof. $M \in R$ and w_j is a unit of $\mathcal{O}$ for every j (for $j = m$ because $\varepsilon \neq 0$), so $1 - cM \in \mathcal{O}$. Its residue in k is

$$\overline{1 - cM} = 1 - c\,\varepsilon^{a_m} \prod_{j \neq m} w_j^{a_j}.$$

The w_j with $j \neq m$ are algebraically independent over $\mathbb{C}(q)$, so this element of k vanishes if and only if the monomial $\prod_{j \neq m} w_j^{a_j}$ is the constant monomial and the scalar is 1: that is, $a_j = 0$ for $j \neq m$ and $c\varepsilon^{a_m} = 1$. This is (i).

For (ii), the hypotheses of (i) give $1 - cM = 1 - cw_m^a$ with $c = \varepsilon^{-a}$. Suppose first $a > 0$. Then

$$1 - \varepsilon^{-a} w_m^a = -\varepsilon^{-a}\big(w_m^a - \varepsilon^a\big) = -\varepsilon^{-a}\,(w_m - \varepsilon) \sum_{i=0}^{a-1} w_m^i \varepsilon^{a-1-i},$$

and the sum has residue $a\,\varepsilon^{a-1}$, which is nonzero in k because $\varepsilon \neq 0$ and $a \neq 0$ in $\mathbb{C}$. Hence t divides $1 - cM$ exactly once. If $a < 0$, write $1 - \varepsilon^{-a} w_m^a = -\varepsilon^{-a} w_m^a \big(w_m^{-a} - \varepsilon^{-a}\big)$ and apply the case just done to $-a > 0$, the prefactor $-\varepsilon^{-a} w_m^a$ being a unit of $\mathcal{O}$. □

Remark 4 (where "exactly one" can fail, and it is not where I expected). The cofactor in 3(ii) is $a\varepsilon^{a-1}$, so "order *exactly* one" is precisely the statement that $a \neq 0$ *in the base field*. Over $\mathbb{C}(q)$ this is free. It is worth recording that it is not free in general:

$$\frac{1 - v^{-1}}{v - 1} = \frac{1}{v} \xrightarrow{v=1} 1, \qquad \frac{1 - v^{-2}}{v - 1} = \frac{v + 1}{v^2} \xrightarrow{v=1} 2.$$

So the type-B half of this step is *characteristic-free*, while the type-C half needs char $\neq 2$: in characteristic 2, $1 - v^{-2} = (1 - v^{-1})^2$ and the type-C boundary factor has order 2. The type that was already proved is the fragile one.

4 The two blocks

Lemma 5 (the ascent block consists of units)**.** *For every* $\alpha \in T(\lambda)$, $\operatorname{ord}_S\big(1 - q\,e^{\alpha}\big) = 0$, *provided* $q \neq \varepsilon^{-h}$. *If* $m \notin \mathbf{I}$ *the hypothesis* $q \neq \varepsilon^{-h}$ *is needed and used exactly once; if* $m \in \mathbf{I}$ *it is not used at all.*

Proof. Apply Lemma 3(i) with $c = q$ to each of the three families of (2).

- $\alpha = t_k \pm t_j$ with $j < k$: the exponent vector is supported on the two *distinct* indices $j \neq k$, so at least one index $\neq m$ has a nonzero exponent. Criterion (i) fails; the factor is a unit. No hypothesis used.
- $\alpha = ht_k$ with $k \notin \mathbf{I}$, $k \neq m$: supported on $k \neq m$; a unit. No hypothesis used.
- $\alpha = ht_m$, which by (2) occurs if and only if $m \notin \mathbf{I}$: the residue is $1 - q\varepsilon^h$, which is nonzero precisely when $q \neq \varepsilon^{-h}$.

In F, q is transcendental over $\mathbb{C}$ and $\varepsilon^{-h} \in \mathbb{C}$, so $q \neq \varepsilon^{-h}$ holds automatically; the point of stating it is the last bullet, which is the *only* place in the whole argument where a condition on q appears, and it appears only when $m \notin \mathbf{I}$. □

Lemma 6 (the inversion block has at most one non-unit)**.** *Let $\alpha \in \mathrm{Inv}(\lambda)$. Then* $\mathrm{ord}_S(1-e^{-\alpha}) = 0$ *unless $\alpha = h\,t_m$, which by* (1) *occurs if and only if $m \in \mathbf{I}$. In that case $1 - e^{-\alpha} = 1 - w_m^{-h}$ and*

$$\mathrm{ord}_S\big(1 - w_m^{-h}\big) = \begin{cases} 1, & \varepsilon^h = 1, \\ 0, & \varepsilon^h \neq 1. \end{cases}$$

Proof. The three families of (1) again. For $\alpha = t_j + t_k$ ($j < k$, both in $\mathbf{I}$) and $\alpha = t_k - t_j$ ($j < k$, $j \notin \mathbf{I}$, $k \in \mathbf{I}$) the exponent vector of $e^{-\alpha}$ is supported on two distinct indices, so Lemma 3(i) fails and the factor is a unit. For $\alpha = ht_k$ with $k \in \mathbf{I}$: $e^{-\alpha} = w_k^{-h}$ is supported on k alone, so (i) requires $k = m$ and $\varepsilon^{-h} = 1$. When $k = m$ and $\varepsilon^h = 1$, part (ii) applies with $a = -h \neq 0$ and gives order exactly 1; when $\varepsilon^h \neq 1$ the residue $1 - \varepsilon^{-h}$ is a nonzero scalar. □

5 The theorem

Theorem 7 (boundary diagonal, uniform in type)**.** *Let $X \in \{B, C\}$, $N \geq 1$, $\mathbf{I} \subseteq [N]$, $m \in [N]$, and $\varepsilon \in \mathbb{C}^\times$. Then*

$$\mathrm{ord}_{S^\varepsilon_m}\, \widetilde{A}^X_{\mathbf{I},\mathbf{I}} \;=\; [\,m \in \mathbf{I}\,] \cdot \#\big\{\, a > 0 \;:\; a\,t_m \in \Delta^+(X_N) \text{ and } \varepsilon^a = 1 \,\big\}. \tag{4}$$

In particular, if $\varepsilon^h = 1$ then $\mathrm{ord}_S\, \widetilde{A}^X_{\mathbf{I},\mathbf{I}} = [\,m \in \mathbf{I}\,]$*, which is* (BD)*; and this holds in* ***type*** *B* ***for all*** *N* ***and all*** $\mathbf{I}$*, closing* G3*.*

Proof. By (3) and additivity of the valuation,

$$\mathrm{ord}_S\, \widetilde{A}^X_{\mathbf{I},\mathbf{I}} = \sum_{\alpha \in \mathrm{Inv}(\lambda)} \mathrm{ord}_S\big(1 - e^{-\alpha}\big) + \sum_{\alpha \in T(\lambda)} \mathrm{ord}_S\big(1 - qe^{\alpha}\big),$$

every summand being defined because every factor lies in $\mathcal{O}$ (Lemma 3). By Lemma 5 the second sum is 0. By Lemma 6 the first sum has at most one nonzero summand, contributed by $\alpha = ht_m$, which belongs to $\mathrm{Inv}(\lambda)$ iff $m \in \mathbf{I}$ and then contributes 1 iff $\varepsilon^h = 1$. Since $\Delta^+(X_N)$ contains exactly one positive multiple of t_m, namely ht_m, the set counted in (4) is $\{h\}$ if $\varepsilon^h = 1$ and $\varnothing$ otherwise, which is the same count. □

Corollary 8 (G4 is derived, not measured)**.** *In type C, $h = 2$ and both $\varepsilon = 1$ and $\varepsilon = -1$ satisfy $\varepsilon^h = 1$: the divisors $\{w_m = 1\}$ and $\{w_m = -1\}$* both *carry* $\mathrm{ord} = [\,m \in \mathbf{I}\,]$*. In type B, $h = 1$, so $\varepsilon^h = 1$ forces $\varepsilon = 1$; on $\{w_m = -1\}$ Theorem 7 gives* $\mathrm{ord}_S\, \widetilde{A}^B_{\mathbf{I},\mathbf{I}} = 0$ *for* every $\mathbf{I}$*, so* (BD) *in the form* $[\,m \in \mathbf{I}\,]$ *fails there, and fails on* precisely *the pairs $(\mathbf{I}, m)$ with $m \in \mathbf{I}$, each by exactly* 1*.*

The 10-19 paper recorded G4 as a measured asymmetry ("$\{w_m = -1\}$ is not a root hyperplane of B_N"), with the measurement taken on *(BC)*. Corollary 8 is the matching statement for *(BD)*, and it is now a consequence of one mechanism rather than two observations: t_m cuts out one hyperplane, $2t_m$ cuts out two.

Corollary 9 ("order one" *is* reducedness)**.** *The order in* (4) *is the number of positive roots proportional to t_m that vanish on S. For the reduced systems B_N and C_N this is at most one. In the non-reduced system BC_N, where t_m* and *$2t_m$ are both positive roots, the same computation returns* $\mathrm{ord}_S\, \widetilde{A}_{\mathbf{I},\mathbf{I}} = 2$ *at $\varepsilon = 1$ and 1 at $\varepsilon = -1$, for every $\mathbf{I} \ni m$.*

Remark 10 (why this matters downstream). (BD) enters the 10-19 paper through $\operatorname{ord}_S \det \Phi = \sum_{\mathbf{I}} \operatorname{ord}_S \widetilde{A}_{\mathbf{I},\mathbf{I}} = \#\{\mathbf{I} \ni m\} = 2^{N-1}$ and through $\operatorname{ord}_S(\Phi^{-1})_{\nu\mathbf{M}} = -1$. Both are *colength* statements, so Corollary 9 says the boundary colength of my lattice is a reducedness count. A non-reduced ambient system would double it. I flag this because the live question Q538 asks whether my object's surviving parity is mod4 rather than mod2 on the grounds that the relevant affine type $A^{(2)}_{2\ell}$ is the *non-reduced* one — and (4) is a place where non-reducedness produces an off-by-a-factor-of-two in a quantity I actually use. I am not claiming the two are the same phenomenon; I am recording that the discriminator is the same (α and 2α both roots) and that here it is now computed, not conjectured.

6 A hypothesis of my own statement, retracted

Remark 11 ((BD) needs no genericity: `rem:generic` is withdrawn). The 10-19 statement of (BD) reads "$\operatorname{ord}_S \widetilde{A}^X_{\mathbf{I},\mathbf{I}} = [\, m \in \mathbf{I}\,]$ for every $\mathbf{I}$, *at a point of S generic in the sense of Remark* `rem:generic`", and that remark explains: $\widetilde{A}_{\mathbf{I},\mathbf{I}}$ contains factors $(1 - qw_m w_k)$ and $(1 - qw_k/w_m)$ which at $w_m = \varepsilon$ become $(1-q\varepsilon w_k)$ and $(1-q\varepsilon^{-1}w_k)$, vanishing on the hypersurfaces $w_k = \pm q^{-1}$; so "generic" was defined to mean *outside* those hypersurfaces.

That hedge is not a hypothesis of the theorem, and I withdraw it. It records a limitation of how I was *measuring*, not a condition on what is *true*. ord_S is the valuation at the generic point of the divisor S (Definition 2): the residue field is $k = \mathbb{C}(q)(w_j : j \neq m)$, in which $1 - q\varepsilon w_k$ is a nonzero element because w_k is an indeterminate. The hypersurfaces $\{w_k = \pm q^{-1}\}$ have codimension 2 in the ambient space and codimension 1 *inside* S; they are proper closed subsets of S and are therefore invisible to a divisorial valuation. They are visible only if one evaluates at a *numerical* point of S and calls the result ord_S — which is what my first $N = 3$ census did when it drew $w_3 = 7/3$ with $q = 3/7$ and reported $\operatorname{ord}_S \det \Phi = 8$ against the correct 4.

The honest description of that episode is therefore not "(BD) requires a generic point" but "my instrument required a generic point, and I wrote the instrument's limitation into the theorem". Every computation in the present paper keeps q and $w_{j\neq m}$ *symbolic* and so sits at the generic point by construction; the hypersurfaces cannot be hit, and no sampling is needed. The replacement hypothesis is the sharp one supplied by Lemma 5: $q \neq \varepsilon^{-h}$, and only on the $m \notin \mathbf{I}$ half.

7 Pre-registration: which arm fired

Written before computing, in `scratch/20261021c2/PREREG.md`; graded arm by arm.

P1 — "(BD) in type B holds for all N, all I, by an argument uniform in type". SPLIT, as predicted, and the split is a result rather than a blemish. The theorem is true and the argument is uniform; what is *not* uniform is the hypothesis class, and the two halves of (BD) turn out to carry **complementary** hypotheses:

half	**needs**	**does not need**
$m \in \mathbf{I}$ (value 1)	$h \neq 0$ in the base field	any condition on q
$m \notin \mathbf{I}$ (value 0)	$q \neq \varepsilon^{-h}$	any condition on h

Neither hypothesis is visible from the other half. The $m \in \mathbf{I}$ half is controlled by $\operatorname{Inv}(\lambda)$ (Lemma 6), the $m \notin \mathbf{I}$ half by the single boundary element of $T(\lambda)$ (Lemma 5), and ht_m lies in

exactly one of the two sets. This is the *second* time in two sessions that a lemma's two halves separated this way (2026-10-21 c1: $k < t$ irremovable in one half of the accounting bijection, inert in the other), and the *third* counting the S/A grouping of 10-19. A binary register cannot record any of them.

P2 — "no genericity hypothesis is needed". TRUE; see Remark 11. This is a retraction of a hypothesis I myself attached to (BD), so it had no natural objector.

P3 — "the order is exactly 1 **because** B_N **and** C_N **are reduced". TRUE**, and the pre-registered falsifier was run: in the non-reduced BC_N the order is 2 at $\varepsilon = 1$ (Corollary 9), confirmed computationally. The same mechanism delivers G4 (Corollary 8).

8 Verification

All computations are pure Python + SymPy (**SageMath is not installed**), with q and $w_{j\neq m}$ kept *symbolic*: ord_S is computed by substituting $w_m \mapsto \varepsilon + t$ and taking the t-adic valuation of the resulting rational function, which is the valuation at the generic point of S. Scripts: `scratch/20261021c2/{g3.py,run_main.py,run2.py,run3.py,run4.py}`.

Control 1 — the transcription. My transcription of the explicit type-B Inv/T products (L1305–1335) agrees with $\widetilde{A}^X_{\mathbf{I},\mathbf{I}}$ computed from Motegi's S_n *symmetrisation definition* — an independent mechanism which never mentions a positive root — for 60/60 pairs $(\mathbf{I}, X)$ at $N \leq 4$, both types. The engine is `scratch/q484_engine.py` carrying the 2026-10-17 $\mathcal{E}^B$ correction (before that date it divided WB by $\mathcal{E}^C$).

Control 2 — the factor list. The list of factors indexed by $\mathrm{Inv}(\lambda) \sqcup T(\lambda)$ multiplies back to the diagonal, 60/60 at $N \leq 4$, both types. So no factor is dropped or double-counted in the bookkeeping that Lemmas 5 and 6 partition.

Control 3 — the instrument is sensitive. Four deliberate mis-transcriptions were run against the engine and *all four were detected*: `h_swap` (type-C boundary root in type B) 10/10; `pair_leq` (non-strict Inv pair product) 10/10; `drop_bdry` 10/10; `T_h_swap` 8/10, the two misses being $\mathbf{I} = [N]$, where $\mathbf{I}^c = \varnothing$ and the plant is a no-op — i.e. correct behaviour, not a failure.

Measurement A — the theorem. $\mathrm{ord}_S \widetilde{A}^X_{\mathbf{I},\mathbf{I}}$ measured from the symmetrisation value (so using neither Lemma 3 nor the factorisation), for all $\mathbf{I}$ and all m:

	$N=1$	$N=2$	$N=3$
type B, $\varepsilon=+1$	2/2	8/8	24/24
type C, $\varepsilon=+1$	2/2	8/8	24/24
type C, $\varepsilon=-1$	2/2	8/8	24/24
type B, $\varepsilon=-1$	1/2	4/8	12/24

The type-B, $\varepsilon = -1$ row is Corollary 8 and its disagreements are *exactly* the $N \cdot 2^{N-1} = 1, 4, 12$ pairs with $m \in \mathbf{I}$, every one measured 0 against a predicted 1. The 34/34 of the top row reproduces the 10-19 G3 number on the same domain.

Measurement B — the general formula (4)**.** 306/306 over nine arms, $X \in \{B, C, BC\} \times \varepsilon \in \{1, -1, i\}$, at $N \leq 3$. The $\varepsilon = i$ arm is a genuine control: there the prediction is 0 everywhere (since $i^1, i^2 \neq 1$), so the formula is not merely fitted to $\varepsilon = \pm 1$.

Measurement C — the reducedness lesion. Non-reduced BC_N, $N \leq 4$: ord $= 2$ on all 49 cases with $m \in \mathbf{I}$ at $\varepsilon = 1$, and ord $= 1$ on all 49 at $\varepsilon = -1$; ord $= 0$ on all 49 cases with $m \notin \mathbf{I}$ in both.

Measurement D — the split lesion. Setting $q := 1 = \varepsilon^{-h}$ at $\varepsilon = 1$, in violation of Lemma 5's hypothesis: the $m \in \mathbf{I}$ half is **inert** ($1 \to 1$, 129/129) and the $m \notin \mathbf{I}$ half **breaks** ($0 \to 1$, 129/129), both types, $N \leq 5$. Corroborated by a second route (full-product expansion rather than factorwise sum), 49/49 each at $N \leq 4$. So the hypothesis is irremovable from one half and inert in the other, which is P1's split exhibited as a lesion rather than argued.

Positive control, named. The pre-registration named $N = 3$, $m = 2$, $\mathbf{I} = \{2, 3\}$, type B, $\varepsilon = 1$ and asserted that all blocks are populated. **That claim was wrong**: at $N = 3$ with $\mathbf{I}^c = \{1\}$ the T pair family (needs $|\mathbf{I}^c| \geq 2$) and the T difference family (needs some $j \notin \mathbf{I}$ above an element of $\mathbf{I}$) are both *empty*, so only 4 of the 6 root families occur. The case still discriminates — it is non-vacuous ($r = 2 \geq 1$, so not the $r = 0$ vacuum, where a green carries zero bits) — but it is weaker than advertised. The smallest case populating *all six* families is $N = 4$, $m = 2$, $\mathbf{I} = \{2, 3\}$, $\mathbf{I}^c = \{1, 4\}$; there the diagonal has 10 factors, exactly one of which ($1 - w_2^{-1}$, from $\alpha = t_2 \in \mathrm{Inv}(\lambda)$) has positive order, and $\mathrm{ord}_S = 1$ from the symmetrisation agrees. The companion $m = 1 \notin \mathbf{I}$ gives 0. Both cases are printed factor-by-factor in `run3.py`/`run4.py` output.

Remark 12 (a fault in my own instrument plumbing, recorded). The trajectory emitter's `log_dir` defaults to the *relative* path `state/trajectory`, so this session's events first landed under `projects/state/traje` with a bare-timestamp session tag — the hazard documented in `emit.py`'s own docstring, which I had read and not acted on. Consolidated to `clio-prove-G3-20261021c2.jsonl`; 18/18 emitted events recorded (a `verify` lacking `verification_target` is silently dropped, so the count is checked, not assumed). `tracecheck` also flagged one of my `verify` events circular, **correctly**: I had offered the transcription-versus-engine comparison as evidence *for the transcription*, and that comparison descends from the transcription. The corrected encoding makes the engine the evidence and the comparison the verify event, and demotes the plant census to what it actually certifies — the *instrument*, not the transcription, since the plants are themselves built from it. Fourth instance of this class; the as-recorded file is kept as `*.asrecorded-circular-encoding`.

9 What this closes, and what remains

G3 — CLOSED. (BD) in type B is Theorem 7. Consequently §§5–7 of `2026-10-19-c2-sa-divisors-full-q.t` — Proposition `prop:lat`, Lemma `lem:inverse`, Lemma `lem:resfun`, Theorem `thm:S` and Theorem `thm:Sprime` — are **proved in both types**, no longer "proved in type C and conditional on G3 in type B". Note the conditionality was on (BD) alone: (BC) in type B is the gauge theorem of `2026-10-19-typeB` and was never in doubt.

G4 — upgraded from measured to derived. Corollary 8, with the exact failure pattern. The 10-19 measurement was of (BC); this is the (BD) half.

G1, G5 — already closed (2026-10-20 c1 and 2026-10-21 c1 respectively).

G2 — still open. Non-cancellation at fixed χ; untouched here, and nothing above bears on it.

A new hypothesis, replacing a withdrawn one. $q \neq \varepsilon^{-h}$, needed only on the $m \notin \mathbf{I}$ half. Automatic in F, but it should be carried in any specialisation of q, and in particular **a $q \to 1$ limit destroys (BD)'s dependence on I entirely** (Measurement D): the order becomes 1 for every m and every $\mathbf{I}$, so the colength count $\#\{\mathbf{I} \ni m\} = 2^{N-1}$ becomes 2^N. Any future classical ($q \to 1$) degeneration of this material must not inherit (BD).

One thing I did not do. I have not re-read Motegi's *proof* of (3) in type B; the paper omits it ("essentially the same as the corresponding type C arguments... we omit these proofs", L1281) and derives it at L3797 from `relationwithAnormalizedtypeB` plus `RTVtheorem`(ii). What I have done is take (3) as the statement to be used — it is a theorem of his paper, read first-hand at L1386 and with its explicit expansion read first-hand at L1305–1335 — and verify that expansion against his *definition* by an independent mechanism (60/60, three controls, four plants). So the dependency here is Motegi's type-B diagonal *theorem*, corroborated against his *definition*, and **not** re-derived from the RTV stable-envelope input. That input is the same unverified root recorded as G5 of the 10-19 paper and now tracked as a registry node; this paper does not reduce it, and I am registering (3) as a premise node with its own grade rather than letting a sentence in this paragraph carry it.